TSM 099 Arithmetic Test - Practice 3

Version 202610122020.18 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Simplify the square root.

$\sqrt{20}$

2. Substitute a = 2, b = -5, c = 2 into the formula. Evaluate both branches.

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

3. Complete the table by evaluating each side separately. State what the tested values establish; do not solve the equation.

$\frac{x^2 - 16}{x - 4} = \frac{x^2 + 10x + 25}{x + 5}$

x	Left value	Right value	Equal?
8			
-8			
0			

4. Reduce each fraction to lowest terms.

(*a*) $\frac{21}{35}$

(*b*) $\frac{70}{110}$

5. Evaluate the principal square root in each part.

(*a*) $\sqrt{49}$

(*b*) $\sqrt{-81}$

Student answer sheet

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Question 1

Radical

20 pts

Question 2

Both values

20 pts

Question 3

x = 8

4 pts

x = -8

4 pts

x = 0

4 pts

Conclusion

8 pts

Question 4

(a)

10 pts

(b)

10 pts

Question 5

(a)

10 pts

(b)

10 pts

Answer key

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Question 1

Radical	$2\sqrt{5}$	20 pts

Question 2

Both values	$x = \frac{1}{2},\ 2$	20 pts

Question 3

x = 8	Left 12; right 13; no	4 pts
x = -8	Left -4; right -3; no	4 pts
x = 0	Left 4; right 5; no	4 pts
Conclusion	None of these values works; this alone does not prove no solution.	8 pts

Question 4

(a)	$\frac{3}{5}$	10 pts
(b)	$\frac{7}{11}$	10 pts

Question 5

(a)	7	10 pts
(b)	9i	10 pts